

Scalability & Future Plan

Date	12 August 2026
Name	Chahat Chouhan
Team ID	Individual Project
Project Name	AI-Brand-Voice-Generator
Maximum Marks	1 Marks

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single API Key & Rate Limits	Concurrent user requests could trigger Groq API rate limits or slow down response delivery.	High
2	Lack of Content History / Persistence	Generated captions, hashtags, and CTAs are lost once the session ends due to no database connection.	Medium
3	Text-Only Generation	System is restricted to generating text outputs without multi-modal image or video synthesis.	Low

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single synchronous Flask server processing one request flow at a time.	Deploy app with Gunicorn + Nginx and implement asynchronous task queues (Celery/Redis).
2	Data Storage	Ephemeral in-memory handling of user inputs and generated results.	Integrate a relational/NoSQL database (PostgreSQL / MongoDB) to save user profiles and past post history.
3	Performance	Dependent on real-time Groq API response latencies	Implement Redis Caching for frequently generated prompt topics and popular hashtag lookup outputs.
4	Security	Basic environment variable API key management (.env).	Upgrade to secret management services (AWS Secrets Manager) and implement OAuth 2.0 / JWT User Authentication .

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	User Accounts & History Dashboard	Q2 2026	Allows users to log in, save favorite captions, track history, and export content drafts.
Phase 3	Direct Social Media Publishing Integration	Q3 2026	Enables one-click posting/scheduling to platforms like Instagram, LinkedIn, and X (Twitter).
Phase 4	Multimodal AI Content Generation	Q4 2026	Expands platform capabilities to generate visual post graphics and video scripts alongside text copy.